

Automated minus reference counts on the gold standard: protocol change

tranche mixed_gold_01 - frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square root axes - every asserted gold row

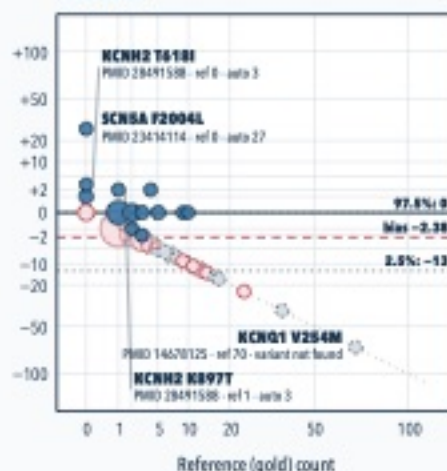
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

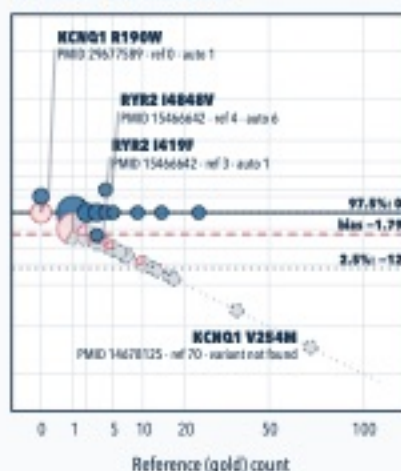
Baseline arm - 20260903_protocol_mixed01_baseline - runtime a7de3089 - locked 2026-09-03 - 49 gene-paper attempts

Carriers



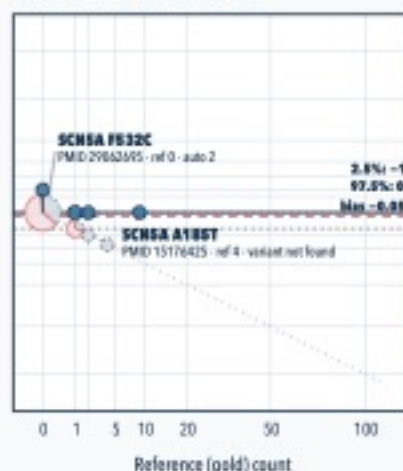
242 gold rows - matched 64.0% - supplied 19.8%
exact 18.6% (zero-baseline 2.1%) - within ± 1 65.7%
bias -2.38 - c2e MAE 2.682 - cond. MAE 0.812 (n 48)
empirical 95% limits -13 to 0 - RMSE 6.60

Affected individuals



242 gold rows - matched 64.0% - supplied 41.7%
exact 45.9% (zero-baseline 6.6%) - within ± 1 76.4%
bias -1.79 - c2e MAE 1.835 - cond. MAE 0.079 (n 101)
empirical 95% limits -12 to 0 - RMSE 5.88

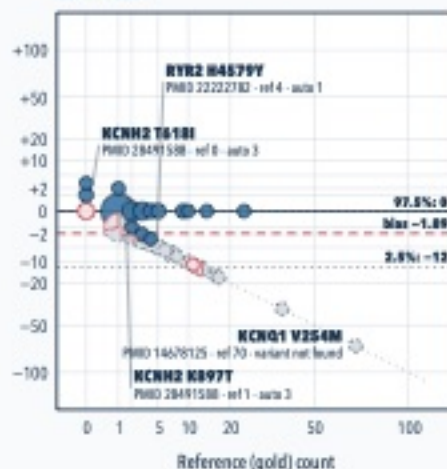
Unaffected individuals



242 gold rows - matched 64.0% - supplied 2.9%
exact 55.9% (zero-baseline 93.4%) - within ± 1 98.8%
bias -0.05 - c2e MAE 0.062 - cond. MAE 0.286 (n 7)
empirical 95% limits -1 to 0 - RMSE 0.36

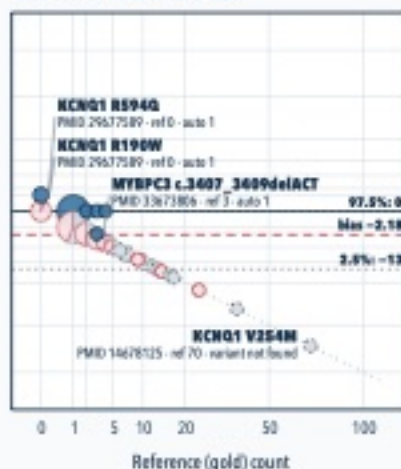
Candidate arm - 20260903_protocol_mixed01_candidate - runtime 4ad7422d - locked 2026-09-03 - 49 gene-paper attempts

Carriers



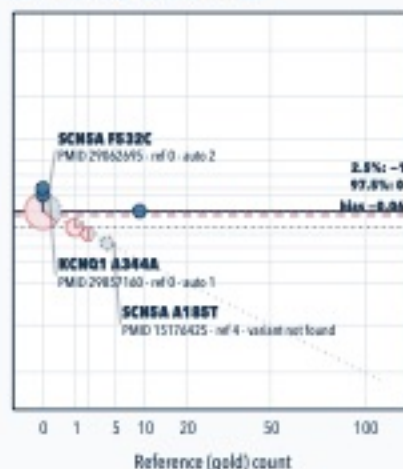
242 gold rows - matched 64.9% - supplied 51.7%
exact 51.2% (zero-baseline 2.5%) - within ± 1 73.1%
bias -1.89 - c2e MAE 1.950 - cond. MAE 0.104 (n 125)
empirical 95% limits -12 to 0 - RMSE 6.02

Affected individuals



242 gold rows - matched 64.9% - supplied 33.5%
exact 38.0% (zero-baseline 6.6%) - within ± 1 68.0%
bias -2.18 - c2e MAE 2.211 - cond. MAE 0.074 (n 81)
empirical 95% limits -13 to 0 - RMSE 6.22

Unaffected individuals



242 gold rows - matched 64.9% - supplied 1.7%
exact 54.2% (zero-baseline 93.4%) - within ± 1 98.3%
bias -0.06 - c2e MAE 0.083 - cond. MAE 0.750 (n 4)
empirical 95% limits -1 to 0 - RMSE 0.40

Candidate - baseline on the same 726 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 2.682 $\rightarrow$ 1.950 (-0.731) - registered UB -0.017
supplied 48 $\rightarrow$ 125 - exact 45 $\rightarrow$ 124 - bias -2.38 $\rightarrow$ -1.89
rows better 86 - worse 5 - unchanged 151
coverage on matched 31.0% $\rightarrow$ 79.8% (+48.65 pp) - UB -1.72 pp

Affected individuals

c2e MAE 1.835 $\rightarrow$ 2.211 (+0.376)
supplied 101 $\rightarrow$ 81 - exact 111 $\rightarrow$ 92 - bias -1.79 $\rightarrow$ -2.18
rows better 6 - worse 25 - unchanged 211
coverage on matched 65.2% $\rightarrow$ 51.6% (-13.57 pp)

Unaffected individuals

c2e MAE 0.062 $\rightarrow$ 0.083 (+0.021)
supplied 7 $\rightarrow$ 4 - exact 232 $\rightarrow$ 228 - bias -0.05 $\rightarrow$ -0.06
rows better 0 - worse 4 - unchanged 238
coverage on matched 4.5% $\rightarrow$ 2.5% (-1.97 pp)

Identity (primary endpoint): recall 64.05% $\rightarrow$ 64.88% (+0.83 pp) - precision 71.76% $\rightarrow$ 74.41% (+2.65 pp) - registered decision: reject_or_revise_candidate - secondary counts